

Customer Shopping Behavior Analysis

A deep dive into **3,900 transactions** across product categories — uncovering spending patterns, customer segments, and subscription behavior to drive smarter business decisions.

TRANSACTIONAL ANALYTICS

PYTHON · SQL · POWER BI



Dataset at a Glance

3,900

Transactions

Rows of customer purchase records

18

Features

Demographics, purchase details, and behavior signals

\$59.76

Avg. Purchase

Mean transaction value across all customers

3.75

Avg. Rating

Average customer review score out of 5

Key data dimensions include customer demographics (age, gender, location), purchase details (category, amount, season), and behavioral signals (discount use, shipping preference, review rating).

Data Preparation in Python

01

Load & Explore

Imported data with `pandas`; used `df.info()` and `.describe()` for initial profiling.

02

Clean & Impute

Imputed 37 missing Review Rating values using per-category median; standardized columns to `snake_case`.

03

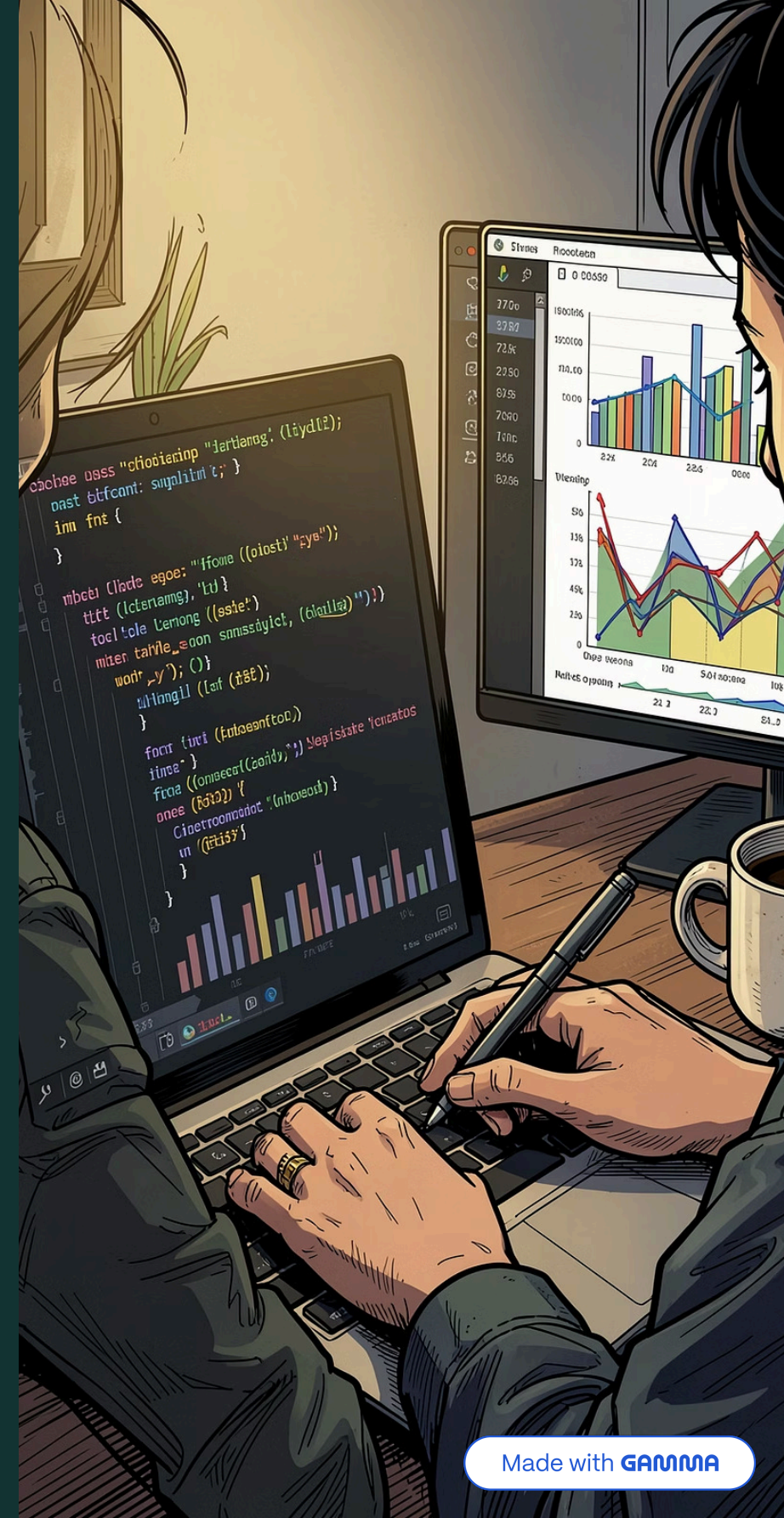
Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used` column.

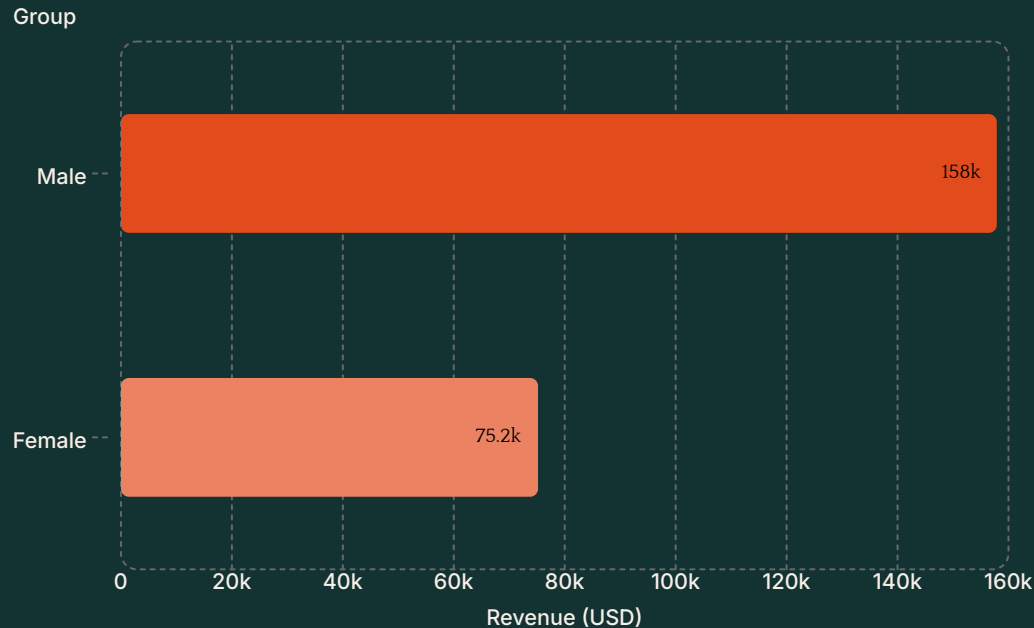
04

Load to PostgreSQL

Connected Python to PostgreSQL and exported the cleaned DataFrame for structured SQL analysis.



Revenue & Spending Patterns



Revenue by Gender

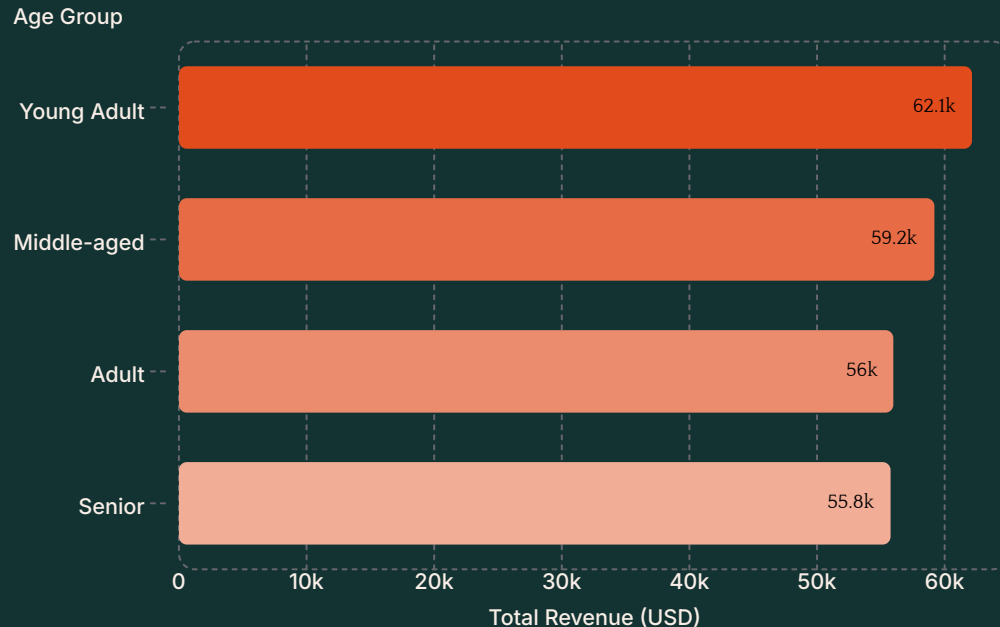
Male customers generated **\$157,890** — more than double the female segment at \$75,191. This disparity likely reflects the higher male representation in the dataset (2,652 vs. 1,248).

Shipping & Spend

Express shipping users spend **\$60.48** on average vs. **\$58.46** for Standard — a signal that express customers may be higher-intent, higher-value buyers worth targeting.

SQL Analysis

Revenue by Age Group



Key Insight

Young Adults lead revenue contribution at \$62,143, followed closely by Middle-aged customers at \$59,197. Revenue is relatively balanced across age groups — suggesting broad appeal but leaving room for targeted campaigns to amplify high-value segments.



Average customer age is **44 years**, ranging from 18 to 70.

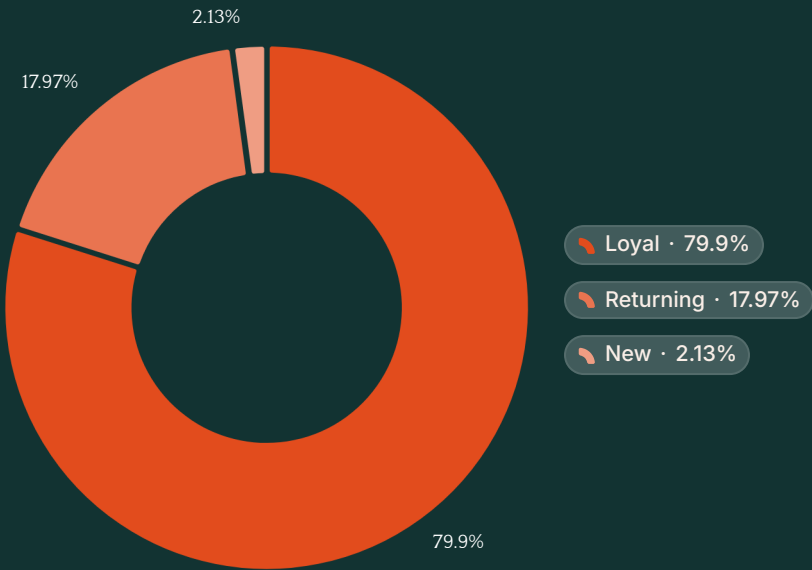
Subscriptions & Customer Segments

Subscription Impact

Status	Customers	Avg. Spend	Total Revenue
Subscribed	1,053	\$59.49	\$62,645
Non-Subscribed	2,847	\$59.87	\$170,436

Subscribers and non-subscribers spend nearly the same per transaction. Non-subscribers drive more total revenue simply due to volume — highlighting a significant opportunity to convert them.

Customer Segmentation



Product Performance

Top-Rated Products

Product	Avg. Rating
Gloves	3.86 ★
Sandals	3.84 ★
Boots	3.82 ★
Hat	3.80 ★
Skirt	3.78 ★

Most Discounted Products

Product	Discount Rate
Hat	50.0%
Sneakers	49.7%
Coat	49.1%
Sweater	48.2%
Pants	47.4%

Top 3 Products by Category

Category	#1	#2	#3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

📌 Hat is both a **top-rated** and **most-discounted** product — pricing strategy review recommended.

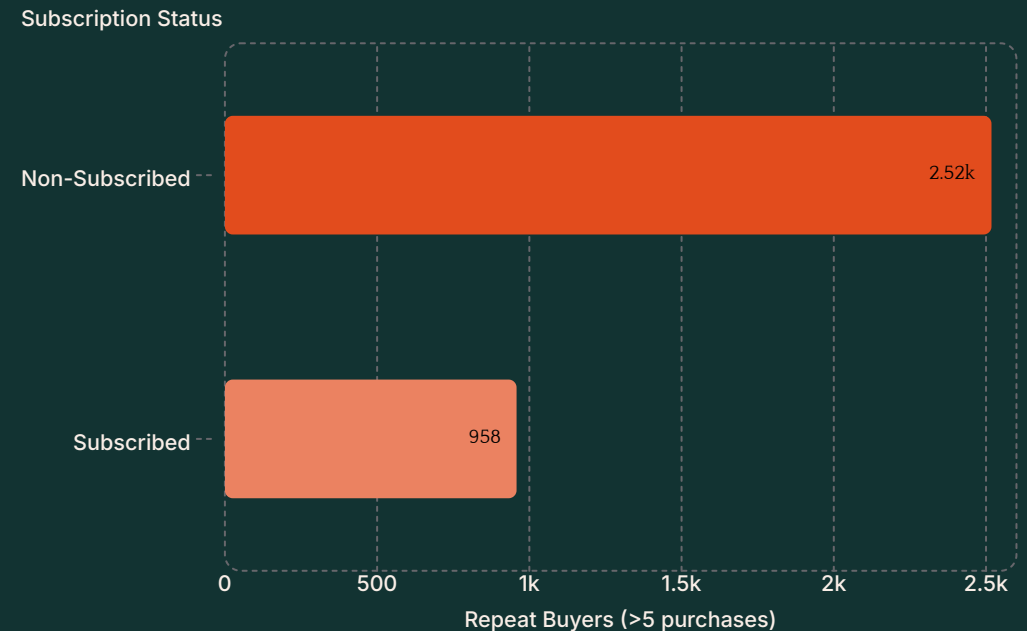
Discount Usage & Repeat Buyers

High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount — demonstrating that discounts attract quality buyers, not just bargain hunters. These customers represent a high-value retention target.

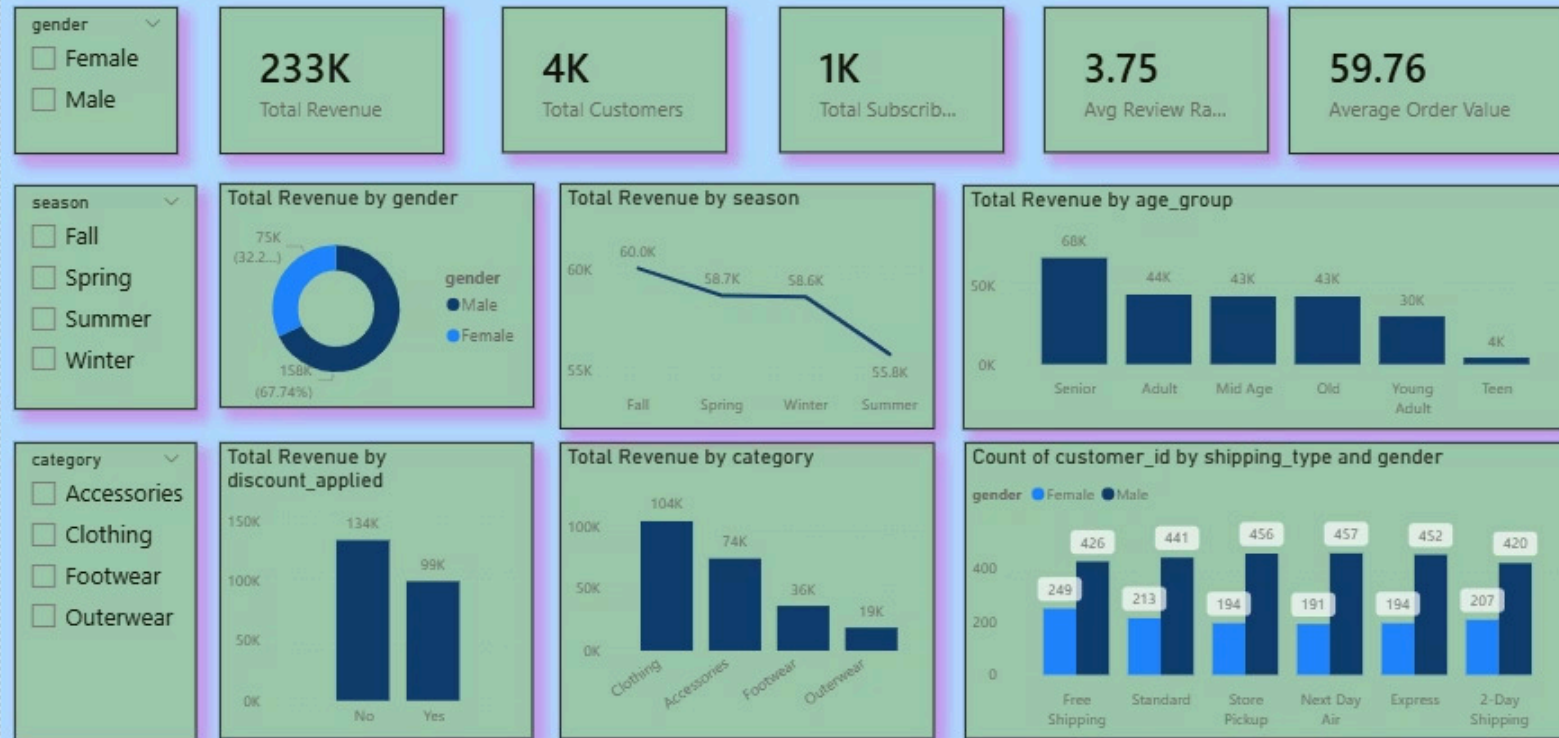
- ✓ 839 out of 3,900 customers (~21%) are high-value discount users spending above the \$59.76 average.

Repeat Buyers & Subscriptions



Even among customers with more than 5 purchases, the majority are **not subscribed** — a clear conversion opportunity.

Customer Shopping Behavior Analysis



Power BI Dashboard

An interactive dashboard consolidates all insights — featuring KPI cards, donut charts for subscription split, and bar charts for revenue and sales by category and age group. Sliceable by gender, category, subscription status, and shipping type.

Strategic Recommendations

Turning Insights into Action



Boost Subscriptions

Only 27% of customers subscribe. Promote exclusive perks – early access, free shipping – to convert the large pool of repeat, non-subscribed buyers.



Reward Loyalty

80% of customers are already "Loyal." A tiered loyalty program can deepen engagement and reduce churn risk among your most valuable segment.



Revisit Discount Policy

Products like Hat and Sneakers see -50% discount rates. Recalibrate to protect margins while maintaining conversion – particularly for already top-rated items.



Targeted Campaigns

Focus marketing on Young Adults and Express shipping users – two segments showing elevated revenue potential and purchase intent.